Maxwell-Fluid Shrinking-Sheet PINN: Reproduction, Evaluator Harness, GPU Candidate Search and Stability-Table Audit

Complete record of the work in the `nnip/` repository

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Summary compiled 12 November 2026
Work performed 5–6 November 2026 (three sessions)

Abstract

This document is a complete, self-contained summary of the work stored in the `nnip/` repository. The subject is a Physics-Informed Neural Network (PINN) reproduction of Pandey *et al.*, *"Insight of thermal transport in Maxwell fluid flow on a porous contracting flat surface inside a porous material with non-uniform heat source/sink: Multiple solutions and stability treatment"*, *Forces in Mechanics* 100352 (DOI 10.1016/j.finmec.2026.100352).

Three things were produced. **(i) A reproduction harness:** the paper's equations, boundary conditions and Tables 1–3 were transcribed into a durable provenance record, and a candidate-proof *external evaluator* was written that gates a PINN script against the paper's published numbers *and* two independent numerical oracles. The baseline PINN was hash-locked twice (CPU and GPU) through the Open Discovery `program-evolution` contract. **(ii) A GPU candidate search:** five candidates were proposed and evaluated under a pre-registered acceptance rule; the winner (candidate-005, a combination of four orthogonal mechanisms) is **2.72× faster end-to-end on the same GPU** with accuracy unchanged at the search point, and was subsequently shown **accuracy-equivalent** to the baseline on unseen parameter rows and seeds. **(iii) A stability-table audit:** Tables 1 and 2 and the critical suction parameter S_c reproduce; **Table 3's eigenvalue magnitudes do not**. Three independent solvers of the paper's own Eq. (31) agree with each other to $< 10^{-7}$ relative and disagree with the published lower-branch column by factors of 2.8–13×, while the *signs* (upper stable, lower unstable) do reproduce. A plausible mechanical explanation for the published upper column is identified: the bundled Chebyshev oracle applies its boundary rows at swapped ends, and that placement reproduces all five published upper values to within 0.7 %. Section 9 weighs the whole record against the article itself: superseded on the stability table, improved on the far-field reference, matched on Tables 1 and 2, and still well behind on computational cost and parameter coverage.

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1 Scope, inputs and provenance

1.1 What the human supplied

A single chat session provided four artifacts, all originating outside the repository:

1. The paper PDF (12 pages, uncorrected proof), `pagination_FINMEC_100352-1.pdf`.
2. `pinn_maxwell_shrinking.py` — a three-stage PINN solving the paper's similarity ODEs (branch solve, fold tracing, stability eigenvalue).
3. `reference_bvp.py` — an independent `scipy.solve_bvp` collocation solver for the same boundary-value problem, used as ground truth.
4. `ref_eig_cheb.py` — an independent Chebyshev-spectral solve of the linearised stability eigenvalue problem, a second ground truth.

The instruction was to use the existing Open Discovery harness (`paper-implementer` for the reproduction claim, `program-evolution` for an optimisation baseline) to give the work a durable, evidence-preserving home.

1.2 Non-destructive provenance

Per the harness contract, external work is never modified. The scripts and PDF were copied read-only into the project and their SHA-256 digests recorded (Table 1). No upstream Open Discovery file was edited: a check of the working tree against the upstream history (`open-discovery/.git-upstream`) shows *zero* modified tracked files. All new material lives in two directories that the harness `.gitignore` deliberately excludes and that were force-added to the outer repository: `initiatives/maxwell-shrinking-pinn/` and `skill-lab/evolve-maxwell-pinn-gpu/`.

Table 1: Source-material SHA-256 digests recorded at copy time (truncated).

File	SHA-256 (prefix)
`pinn_maxwell_shrinking.py`	`8f145d68377b7be4bbbc73f...b180`
`reference_bvp.py`	`a1e042fd16bdccd19cdac98...f203`
`ref_eig_cheb.py`	`081318bf7cbb5ab4a78d7e9...60b1`
`pagination_FINMEC_100352-1.pdf`	`005acd8e27c6533be8e9e4e...f5e3`
Locked baseline bundle (`lock.json` root)	`40d7e8611b8d417153eea25...c281`
Evaluator (`evaluator.py`)	`e99f0be74b50d55d74e5e8a...40eb`
Candidate-005 winner, rebuilt from patch	`cae54498...0661`

1.3 Repository layout

- `nnip/amit-pinn-article/` — the original source material (PDF, three scripts) plus `requirements-gpu-cu126.txt`, the exact pinned GPU environment (torch 2.14.1+cu126, scipy 1.18.1, numpy 2.5.4). The 6.7 GB venv itself is excluded from version control and rebuilt from this file.
- `.../open-discovery/` — the research harness (upstream, unmodified): `AGENTS.md`, 20 workflow skills under `.agents/skills/`, seven agent role definitions, `docs/` (evidence standard, candidate lifecycle, state model), and the `program-evolution/` locking harness.
- `.../initiatives/maxwell-shrinking-pinn/` — **all new scientific work** (2.2 MB): `BRIEF.md` and the single project `projects/pinn-reproduction/`.
- `.../skill-lab/evolve-maxwell-pinn-gpu/` — **new** prototype skill: how to run a GPU candidate search for this specific initiative, plus a menu of concrete candidate mechanisms.

2 The physical problem and the claims under test

2.1 Governing similarity ODEs (paper Eqs. 13–15)

Two-dimensional steady boundary-layer flow and heat transfer of a Maxwell (viscoelastic) fluid over a continuously *contracting* sheet embedded in a Darcy porous medium, with wall mass transfer and a heat source/sink depending on *both* temperature and spatial position:

$$(1-\beta f^2)\, f''' + 2\beta f f' f'' + f f'' - f'^2 - K f' = 0, \tag{1}$$

$$\theta'' + \Pr f\, \theta' + A^* \theta + B^* e^{-\eta} = 0, \tag{2}$$

subject to

$$f(0) = S, \quad f'(0) = -1, \quad f'(\infty) = 0, \qquad \theta(0) = 1, \quad \theta(\infty) = 0. \tag{3}$$

Here β is the Deborah (Maxwell) number, K the permeability parameter, S the wall mass-transfer parameter ($S > 0$ suction), Pr the Prandtl number, and A^*, B^* the temperature- and space-dependent source/sink strengths. The paper truncates the far field at $\eta_\infty = 10$. The engineering outputs are

$$C_f \operatorname{Re}_x^{1/2} = (1+\beta) f''(0), \qquad \operatorname{Nu} \operatorname{Re}_x^{-1/2} = -\theta'(0). \tag{4}$$

2.2 The paper's own method, and why cross-validation matters

The paper solves the system by shooting (RK4, $\Delta\eta = 0.01$, Secant iteration to 10^{-5}). Neither companion script uses that route: `reference_bvp.py` uses collocation and the PINN uses a neural trial function with hard-constrained boundary conditions. Three independent numerical routes to the same claim is exactly the right design for an audit, and this independence is what later made the Table 3 discrepancy diagnosable rather than ambiguous.

2.3 Claims transcribed as automated gates

`paper/paper-summary.md` transcribes the equations, boundary conditions, solution method, all of Tables 1–3, and the qualitative conclusions of Section 6. Each number is explicitly marked either **[gate]** (used as an automated evaluator gate, cross-checked against the co-author's own inline comments in the original script) or **[reference only — verify against PDF before quoting]** (read visually off rendered table images). The gated values are listed in Table 2.

Table 2: The published values used as automated evaluator gates. Baseline parameter row: $\beta = 0.1$, $K = 0.1$, $S = 1.95$, $\Pr = 2$, $A^* = B^* = 0.1$; stability row $S = 2.0$, $B^* = 0$.

Claim	Source	Upper branch	Lower branch
$C_f \operatorname{Re}^{1/2}$	Table 1, p. 7	1.93523180	0.75244081
$\operatorname{Nu} \operatorname{Re}^{-1/2}$	Table 2, p. 7	3.35807824	3.14427924
Critical suction S_c	Sec. 4.1, Fig. 2	1.8391	
Leading eigenvalue α_1	Table 3, p. 10	+0.8810518	−0.0376726

The physics being protected: dual (upper/lower) similarity solutions exist only for $S \geq S_c$; the upper branch is stable ($\alpha_1 > 0$) and the lower branch unstable ($\alpha_1 < 0$) at every tabulated S. A "faster" candidate that converges to one branch, swaps branch labels, or flips an eigenvalue sign has improved nothing, however low its wall-clock.

3 The baseline program

`baseline/pinn_maxwell_shrinking.py` (290 lines, float64 throughout, device auto-selected as CUDA when available) implements three separate PINNs:

1. `solve_branch(p, branch, ...)` — the upper or lower branch at one parameter point. A 4×40 tanh MLP (Xavier-normal init, zero bias) on the mapped variable $\xi = 1 - e^{-\eta/L}$ with 300 collocation points up to $\xi_{\max} = 0.9995$. Far-field boundary conditions are imposed *exactly* by the trial function, not by penalty. Training is 4000 Adam (lr $= 2 \times 10^{-3}$) + 1500 LBFGS for momentum, then $2000 + 1000$ for energy, preceded by a $2500 + 200$ "pilot" solve whose only purpose is to read off $f(\infty)$ and choose the map length L_{MAP} (`auto_scale=True`). Branch selection uses an `f2_anchor` warm start.

2. `trace_fold(p, f2_values)` — treats S as the unknown and prescribes $f''(0)$ over eight continuation values, so the solution path passes smoothly through the turning point and yields S_c directly.

3. `smallest_eigen(base_net, p, alpha_init)` — the leading eigenvalue α_1 of the linearised disturbance problem about a frozen base flow ($3000 + 1000$ iterations, α a trainable parameter).

The script's own docstring records that L_{MAP} auto-scaling is what brings the lower branch from a 1–2 % error floor down to ∼0.1 %; this is flagged in the project notes as something a candidate must not silently remove.

4 Infrastructure built

4.1 The external evaluator (`evaluator.py`, 364 lines)

A standalone, *import-based* (not print-parsing) evaluator implementing the `program-evolution` contract: it exits 0 and prints exactly one JSON object. It loads a candidate module via `importlib` and first checks the required public contract `{MLP, train, fields, wall_values, solve_branch, trace_fold, smallest_eigen}`. A candidate may change anything *inside* those functions; it may not change their signatures, and it may never edit the evaluator.

`-mode smoke` **(default, cheap).** Monkeypatches the candidate's `train` to a hard 120-Adam/40-LBFGS budget and checks the dual-solution pipeline runs and returns finite, physically ordered values ($C_f^{\mathrm{up}} > C_f^{\mathrm{low}} > 0$, both Nu > 0). This is a wiring/pipeline gate, explicitly *not* an accuracy claim — the evaluator docstring says so, and the project notes repeat it.

`-mode full` **(expensive).** Runs the candidate's own unmodified `solve_branch` at the paper's baseline row and gates on relative error to the published C_f and Nu (default tolerance 1 %).

`-fold` gates S_c against 1.8391 (tolerance 3 % in the acceptance rule).

`-stability` gates the Table 3 eigenvalues, including a *hard sign check* (upper > 0, lower < 0) — the real guard against branch confusion.

`-oracle` cross-checks the *paper's* numbers against the independent `scipy.solve_bvp` oracle. Informative only; it never gates the candidate.

The objective reported is `elapsed_seconds` (minimised), summed over the stages actually requested.

4.2 Baseline locks

`program-evolution/scripts/lock_baseline.py` snapshots baseline + evaluator with hashes, environment, and a smoke run.
Note that the two locks reproduce the smoke values to 10–11 significant figures across operating system, Python version, torch version and device — a strong cross-platform determinism check that was obtained for free from the locking discipline.

Table 3: The two baseline locks. Both report `status: locked`, `error: null`, `valid: true`, `SUPPORTED`.

	`baseline-lock/` (session 1)	`baseline-lock-gpu/` (session 2)
Hardware	CPU only (Windows 10)	NVIDIA GTX 1080 Ti (sm_61)
torch	2.13.0+cpu	2.14.1+cu126 (driver 580.178.04)
Python / OS	3.13.7 / Windows-10	3.12.15 / Linux 6.8 glibc 2.39
Smoke objective	**126.81 s**	**14.77 s**
Smoke C_f upper / lower	1.9125139387 / 0.7721875369	1.9125139387 / 0.7721875369

4.3 The GPU-search skill and mechanism menu

`skill-lab/evolve-maxwell-pinn-gpu/SKILL.md` is a new repository-local skill. Its substantive content is methodological discipline:

1. Verify CUDA is genuinely present before any speed claim.
2. **Never compare wall-clock across hardware.** Moving to a GPU is a hardware effect, not an evolved program; a fresh same-hardware reference must be established first, and the hardware and algorithmic improvements must be reported separately and *never summed*.
3. One candidate folder per hypothesis, gate on smoke first, spend full-mode budget only on survivors, never edit the evaluator.
4. A claimed winner must be *reconstructed* from the locked baseline plus its recorded patch and re-confirmed on fresh parameter points over several seeds.
5. Novel-science extensions are a *new project*, not a candidate.

`references/candidate-mechanisms.md` is a menu of 14 concrete, line-referenced hypotheses in four groups: **A** same-hardware throughput (A1 batch both branches via `vmap`; A2 cheaper L_{MAP} pilot; A3 early-exit LBFGS; A4 Adam/LBFGS rebalance; A5 vectorise the fold continuation loop; A6 trade collocation count for iteration count), **B** accuracy and robustness (closed-form L_{MAP}, more reliable branch selection, stress-testing the $(1-\beta f^2)$ singularity, targeting the lower branch), **C** GPU-native restructuring under the same public contract, and **D** a novel-work track (parametric/meta-PINN over (S, β, K), continuous fold-tracking by differentiating with respect to S, parametric stability locating $\alpha_1 = 0$ as an independent check that the turning point and the stability-exchange point coincide, and ensemble uncertainty).

4.4 Independent confirmation and diagnostic tooling (new code)

- `confirm_fresh.py` (55 lines) — runs a program's own `solve_branch` at three Table 1 rows *not* used during the search (S=2.10; K=0.13; β=0.15) over several seeds, and compares C_f and Nu both with the paper and with a long-domain (η_∞=40, converged) `scipy.solve_bvp` oracle.
- `diagnostics/eig_shooting.py` — independent IVP shooting solve of the paper's Eq. (31), with a domain-length study.
- `diagnostics/eig_cheb_fixed.py` — the bundled Chebyshev oracle with its boundary rows moved to the correct ends.
- `oracle-v2/table3_oracle.py` (139 lines) + `judge.py` — a full Table 3 oracle (three methods × five domain lengths × five S values × two branches) with the pre-registered acceptance criteria applied mechanically to produce `judge.md`.
- `summarize_run.py`, `run_candidates.sh`, `queue_round2.sh`, per-run `run.sh`/`timeline.txt`/`env.txt`/`nvidia-smi.txt`/ `sha256.txt` — the evidence plumbing.

5 Session 1 (5 Nov 2026, CPU): harness bring-up

- Paper read in full (12 pages); equations, boundary conditions, solution method and Tables 1–3 transcribed with gate/reference-only marking.
- Source material copied read-only with hashes recorded; originals untouched.
- `evaluator.py` written and validated in smoke mode against the baseline: **valid, SUPPORTED**, $C_f^{\text{up}}\approx1.913$, $C_f^{\text{low}}\approx0.772$, $\text{Nu}^{\text{up}}\approx1.854$, $\text{Nu}^{\text{low}}\approx1.203$ in 126.8 s — correctly ordered and finite, and explicitly *not* claimed to match the paper.
- Baseline locked (`baseline-lock/lock.json`, `status: locked`).
- A full-mode CPU confirmation run was launched in the background. **It never completed** — `runs/full_table12_confirmation.json` is empty, and this is recorded as such rather than quietly dropped. The notes stated in advance that no pass/fail may be reported before the process exits.

6 Session 2 (5 Nov 2026, GTX 1080 Ti): GPU reference and candidate search

6.1 Pre-registration, written before any candidate result

`runs/GPU-SEARCH-PREREGISTRATION.md` fixes the acceptance rule *before* results exist:

1. smoke gate valid;
2. full mode: no Table 1/2 failure (tol 1 %), no S_c failure (tol 3 %);
3. stability: both sign checks pass; α_1 magnitudes *reported* against the independent values, not used to accept or reject;
4. the speed claim is the same-GPU total objective versus 1451.8 s;
5. a winner must be rebuilt from baseline + recorded patch and re-confirmed on fresh parameter points over several seeds.

Two reasons are documented for excluding the α_1 magnitude gate, both established before candidate results:

- **No correct solver can satisfy it.** Two independent solves of the paper's own Eq. (31), verified term-by-term against page 10, agree with each other and not with Table 3: shooting gives $\alpha_1^{\text{up}} = 0.7358$, $\alpha_1^{\text{low}} = -0.4938$ at S=2.0; the corrected Chebyshev solve gives identical roots. $|\alpha_1^{\text{low}} + 0.0377| < 1\,\%$ is unreachable. The evaluator was *not* edited — the program-evolution contract forbids weakening a gate mid-search; the gate is simply reported.
- **The upper-branch magnitude is not identifiable by this PINN.** In the far field $g' \sim e^{-\mu\eta}$ with $(1-\beta f_\infty^2)\mu^2 - f_\infty\mu + (\alpha - K) = 0$. For $\alpha > K$ both roots have $\text{Re}\,\mu > 0$, so $g'(\infty) = 0$ holds for a *continuum* of α on the semi-infinite domain. The PINN trial function imposes $g'(\infty) = 0$ exactly, so the upper root ($\approx 0.74 > K = 0.1$) is not pinned — CPU tests with initialisations spanning 0.0–1.0 all land in 0.66–0.69, while the GPU baseline lands at 0.958. The lower branch ($\alpha \approx -0.49 < K$) has one growing mode, hence a genuine discrete eigenvalue, and the PINN matches the independent value to $\sim$1 %. Conclusion recorded in advance: *only the upper sign is a meaningful PINN output*, and no candidate targets the upper magnitude.

6.2 The same-GPU reference run

`runs/full_confirmation_gpu.json`: unmodified baseline, unmodified evaluator, `-mode full -oracle -fold -stability`. Total objective **1451.79 s** (Table 1/2 404.54 s, fold 456.02 s, stability 591.23 s).

Table 4: GPU reference run against the paper. Table 1/2 and S_c pass at 1 % and 3 %; Table 3 magnitudes fail; both signs pass.

Quantity	Paper	PINN (GPU)	Rel. error
C_f upper	1.93523180	1.93516109	0.0037 %
C_f lower	0.75244081	0.75562700	0.423 %
Nu upper	3.35807824	3.35003942	0.239 %
Nu lower	3.14427924	3.13232975	0.380 %
S_c	1.8391	1.84910	0.543 %
α_1 upper	+0.8810518	+0.958033	8.74 % (**FAIL**)
α_1 lower	−0.0376726	−0.488198	1195.9 % (**FAIL**)

The `-oracle` cross-check confirms the transcription and the paper's own Table 1/2 numbers independently: `scipy.solve_bvp` returns $C_f^{\text{up}} = 1.93525922$ (1.4×10^{-3} % from the paper), $C_f^{\text{low}} = 0.75243975$ (1.4×10^{-4} %), $\text{Nu}^{\text{up}} = 3.35133372$ (0.20 %), $\text{Nu}^{\text{low}} = 3.13525713$ (0.287 %). In other words the oracle is itself 0.2–0.3 % from the paper's Nu values, so the PINN's 0.24–0.38 % is at the resolution of the reference, not a PINN deficiency.

6.3 Candidate lineage

Five candidates, each a full copy of the baseline with one recorded change (`patch.diff`), one hypothesis (`NOTE.md`), and its own evaluator outputs. All five passed smoke (8.0–11.9 s) and all five passed the pre-registered full-mode gates.

Table 5: Candidate lineage, single runs, same GTX 1080 Ti, unmodified evaluator. The "vs baseline" column is against the 1451.8 s reference.

ID	Parent	Change	Total (s)	Speed-up	Gates
001	baseline	Taylor-mode derivative jets instead of nested autograd	823.5	1.76×	pass
002	baseline	adaptive LBFGS stop (50-iter chunks, stop at < 1 % gain)	967.7	1.50×	pass
003	baseline	collocation $\xi_{\max}$ 0.9995 → 0.99995	1374.9	none (same work)	pass[1]
004	baseline	auto-scale pilot 2500/200 → 800/100	1230.7	∼1.18×[2]	pass
005	001 (+2/3/4)	combination of all four mechanisms	**503.5**	**2.88**×	pass

Candidate-001 (the main mechanism). Every optimiser step of the baseline builds f', f'', f''' (and $\theta', \theta'', g', g'', g'''$) with three nested `torch.autograd.grad(create_graph=True)` passes and then backpropagates through that nested graph. For a 4×40 tanh MLP on 300 points this is almost pure kernel-launch overhead on a GPU. Candidate-001 propagates the value and its first k η-derivatives forward in closed form (Faà di Bruno through `tanh`, Leibniz for the trial-function products), giving an identical residual with a much smaller graph. Architecture, initialisation, seeds, collocation, iteration counts and optimiser settings are unchanged. Verification before any evaluator run: on random weights the jet residuals *and their parameter gradients* match baseline autograd to $\leq 2 \times 10^{-14}$ in float64, and the eigen trial-function jet to $\leq 1.1 \times 10^{-15}$.

Candidate-002. The LBFGS tolerances (10^{-12} gradient, 10^{-15} change) never trigger, so every stage burns its full fixed budget. A CPU trace showed momentum converges in ∼100 LBFGS

[1] Candidate-003's CPU accuracy gain was *not* reproduced in full mode — recorded as a negative result rather than dropped.

[2] Within run-to-run noise; single-run stage timings vary ∼15 % (the fold stage, untouched by 003/004, ran 368–387 s against the reference's 456 s).

evaluations and energy in ∼200; the remaining ∼1700 and ∼1050 evaluations move C_f/Nu by $< 10^{-5}$ relative. The candidate runs the same LBFGS in 50-iteration chunks (history preserved) and stops when a chunk improves the loss by $< 1\,\%$, never exceeding the original budget.

Candidate-003. $\xi_{\max} = 0.9995$ enforces the residual only out to $\eta \approx 7.6L$, while the lower branch's slow decay ($\lambda \approx 0.77$) still matters to $\eta \approx 20$. Raising $\xi_{\max}$ to 0.99995 extends the reach to $\sim 13L$. A CPU pre-check showed lower-C_f error against the $\eta_\infty \to \infty$ oracle falling from 0.19 % to 0.018 % — but this gain did *not* reproduce in full mode, which is recorded explicitly.

Candidate-004. The pilot only needs $f(\infty)$ to choose $L = \lceil\lambda\rceil/\lambda$. A budget sweep (2500/200, 1500/100, 800/100, 400/60) moved the chosen lower-branch L only within 1.3095–1.3111, so the pilot was cut to $800 + 100$.

Candidate-005. The four mechanisms act on different things (derivative evaluation, LBFGS stopping, collocation reach, pilot budget), so the hypothesis was that the gains multiply. Candidate-005 is candidate-001 plus the recorded baseline-relative patches of 002, 003 and 004 applied in order; all hunks applied cleanly. The note records the discipline that it is only to be trusted because each component passed independently in round 1.

6.4 Winner confirmation

The winner was rebuilt from `baseline-lock-gpu/baseline/` plus `candidate-005/patch.diff` and verified byte-identical to the evaluated candidate (SHA-256 `cae54498...0661`). Paired full-mode runs were then taken back-to-back on the same otherwise-idle GPU (17:49–18:44 UTC, timeline recorded).

Table 6: Paired same-GPU full-mode runs (`-oracle -fold -stability`), unmodified evaluator. **Algorithmic improvement: 2.72×.**

Stage	Baseline (s)	Candidate-005 (s)	Speed-up
Table 1/2 (both branches, incl. pilots)	392.8	158.1	2.48×
Fold / S_c	396.4	161.9	2.45×
Stability (2 branch solves + 2 eigen)	600.7	191.7	3.13×
Total	**1389.9**	**511.6**	**2.72×**

The hardware effect is reported separately and *not* summed: the Windows-CPU smoke lock took 126.8 s against the GPU smoke lock's 14.8 s — different machines, smoke mode only.

Table 7: Accuracy at the search point is unchanged within noise.

Metric	Baseline	Candidate-005	Reference
C_f upper	1.935161 (0.004 %)	1.934442 (0.041 %)	paper 1.93523180
C_f lower	0.755627 (0.42 %)	0.755643 (0.43 %)	paper 0.75244081; converged oracle 0.753702
Nu upper / lower error	0.24 % / 0.38 %	0.27 % / 0.39 %	the oracle is itself 0.20 % / 0.29 % from the paper
S_c	1.84910 (0.54 %)	1.84947 (0.56 %)	paper 1.8391
α_1 lower	−0.48820	−0.48808	independent −0.4938 (paper −0.0377)
α_1 upper (sign only)	0.958	1.051	independent leading 0.7358 (paper 0.881)

6.5 Fresh-parameter check and the honest open item

`confirm_fresh.py` was run on candidate-005 at three unseen Table 1 rows over seeds 1–3 (the search used seed 0 and only the S=1.95 and S=2.0 rows). Errors against the converged η_∞=40 oracle: upper C_f 0.03–0.21 %, all Nu $\leq$ 0.34 %, lower C_f 0.04–1.00 %. Against the paper's own Table 1 C_f, 16 of 18 points fall within 1 %; the two exceptions are the lower branch at S=2.1 (seeds 2 and 3: 1.51 %, 1.79 %), where *the paper's own value* is 0.78 % below the converged oracle — an η_∞=10 truncation effect that grows as the lower branch decays more slowly.

The session-2 verdict deliberately left one item open rather than overclaiming: whether the lower-branch seed spread is inherent to the baseline architecture or was *introduced* by candidate-005 was untested, because the standard plan had not run the baseline through `confirm_fresh.py`. That comparison was named as the required next step before calling 005 accuracy-equivalent off the search point.

7 Session 3 (6 Nov 2026, RTX 2060 SUPER): closing both open items

The venv was rebuilt from `requirements-gpu-cu126.txt` (torch 2.14.1+cu126, scipy 1.18.1 — identical to session 2). Because the GPU differs, *no* timing from this session is compared with session-2 timings. Both items were pre-registered in `runs/OPEN-ITEMS-PREREGISTRATION.md` with program hashes verified before running.

7.1 Item A — is the lower-branch seed spread candidate-005's fault?

Pre-registered rule. Run unmodified `confirm_fresh.py` on both programs, seeds 1–3, serially on the same idle GPU (baseline first); 9 (row, seed) points per program. Primary metric: lower-branch C_f error against the η_∞=40 oracle. With m and x the mean and max over the nine points, declare *equivalent* if $|m_5 - m_B| \leq 0.20$ percentage points *and* $x_5 \leq x_B + 0.30$ pp; *005 worse* or *005 better* otherwise. The pre-registration states plainly that $n = 9$ per arm detects only gross differences, that "equivalent" means "no gross difference detected" and not proof of equality, and that the two programs consume the RNG differently so the points are not truly paired.

Result: equivalent on the primary metric and all three secondaries (Table 8). The lower-branch C_f seed spread (0.03–1.0 %) is present in the *untouched baseline*; candidate-005 did not introduce it.

Table 8: Item A, errors against the converged oracle over 9 (row, seed) points each, RTX 2060 SUPER, job pinned to 4 CPU cores, 04:54–08:12 UTC.

Metric	Baseline mean / max (%)	005 mean / max (%)	Verdict
Lower C_f (PRIMARY)	0.459 / 0.962	0.392 / 0.988	equivalent
Upper C_f	0.172 / 0.304	0.101 / 0.223	equivalent
Upper Nu	0.058 / 0.156	0.090 / 0.183	equivalent
Lower Nu	0.147 / 0.333	0.180 / 0.347	equivalent

Side findings (informative). Candidate-005 on this GPU reproduces its session-2 (1080 Ti) values to $\leq$ 0.05 pp at 8 of 9 points; the ninth (S=1.95, β=0.15, seed 1) differs materially (0.21 % against 0.67 %) with the same seed on a different GPU — i.e. GPU-level non-determinism matters at the $\sim$0.5 pp level on the lower branch. Wall-clock over the nine fresh points on this GPU: baseline 7671 s, 005 4148 s, ratio 1.85× (branch solves only, CPU-pinned — explicitly *not* comparable with the 2.72× full-mode figure).

7.2 Item B — building a correct Table 3 oracle

Pre-registered hypothesis. On the semi-infinite domain the disturbance problem has a continuous spectrum for $\alpha > K$; finite-domain solvers then produce roots that accumulate as L grows, and the lowest such root tends to the onset of oscillatory far-field decay,

$$\alpha^* = K + \frac{f_\infty^2}{4\,(1 - \beta f_\infty^2)}, \tag{5}$$

rather than to a discrete eigenvalue. Convergence criteria, the support/refutation conditions, and the rule that the PINN's stability stage is never re-tuned against the new oracle were all fixed in advance.

Deliverable. `oracle-v2/table3_oracle.py` (the bundled `oracle/` stays byte-identical), sweeping three methods — shooting with $g'(L) = 0$, corrected Chebyshev, and shooting with the exact asymptotic boundary condition — over $L \in \{10, 12, 16, 20, 24\}$, $S \in \{2.0, \ldots, 2.4\}$ and both branches; ∼25 minutes on CPU. Eq. (31) was re-checked against the page-10 image: the coded operator is the exact linearisation of steady Eq. (13) with $+\alpha g'$, whereas the *printed* equation drops primes (it shows $+2ff''g_0$ and $-2ff''g_0$, which would cancel), so the linearisation is taken as the authority.

Table 9: Table 3 oracle v2 (β=0.1, K=0.1, A^*=0.1, Pr=2, B^*=0).

S	Lower α_1 (converged)	Paper lower	Upper lowest finite-L root	α^*	Paper upper
2.0	−0.49257	−0.0376726	0.7348 (converged)	0.8205	0.8810518
2.1	−0.61135	−0.1290175	1.0493 (converged)	1.0882	1.1498579
2.2	−0.70803	−0.2006274	∼1.39 (0.36 % drift $L20 \to 24$)	1.4016	1.4609727
2.3	−0.79412	−0.2601093	not converged ($1.79 \to 1.92$)	1.7828	1.8396721
2.4	−0.87670	−0.3115262	not converged ($2.28 \to 2.39$)	2.2618	2.3116364

B1 — the lower branch is converged, and Table 3's lower column does not reproduce. All three methods agree to $< 10^{-7}$ relative at every S, and the value changes by $< 10^{-5}$ between L=20 and L=24; the pre-registered convergence criterion is met at every row. The *sign* agrees with the paper (lower branch unstable), but the magnitudes are 2.8× to 13× larger than published. The discrepancy is therefore on the paper's side, reproduced from the paper's own Eq. (31)/(33).

B2 — the upper-branch hypothesis was refuted, in a way the pre-registration did not foresee. At S=2.0 and 2.1 the lowest finite-L root does *not* approach α^* from above: it is converged ($< 0.02\,\%$ change over L=16–24) and lies *below* α^* ($-10\,\%$, $-4\,\%$). Higher roots do crowd toward α^* as L grows. The literal refutation condition ("converged and $> 2\,\%$ *above* α^*") was not met either — the outcome was a third case, and this is stated as such rather than being retrofitted into the pre-registered dichotomy. Interpretation offered but not further tested: for $K < \alpha < \alpha^*$ both far-field modes decay monotonically, and $g'(L) = 0$ at large L forces the slow-decaying mode's coefficient to vanish, so the converged root is the "fast-decay" eigenvalue; under $g'(\infty) = 0$ alone every $\alpha > K$ still qualifies. At $S \geq 2.2$ neither method converges (Chebyshev becomes ill-conditioned for $L \gtrsim$ 16–20, producing spurious $|\alpha| \sim 10^2$; shooting drifts with L), so **no reliable upper magnitude is claimed for** $S \geq 2.2$. What is robust: every upper candidate root exceeds $K > 0$, so the upper branch is stable — the sign agrees with the paper.

B3 — attribution of the published upper column (informative only). The bundled `oracle/ref_eig_cheb.py` applies the wall boundary conditions at $\eta = L$ and the far-field condition at the wall — the rows are at swapped ends. At its own default L=12 that placement yields upper values 0.88253, 1.14325, 1.45075, 1.82675, 2.30169: within 0.17–0.70 % of *all five* published Table 3 upper values. The same placement gives positive values on the lower branch, so it does

not reproduce the lower column. This is recorded as *consistent with, but not proof of*, the paper's upper column having come from a computation with that boundary placement.

B4 — energy-disturbance eigenvalues (new, Eq. 32 with $g_0 = 0$). Never examined before. All finite-L energy eigenvalues are positive (lower branch 0.03–0.12 at L=24, decreasing with L), so they change neither sign conclusion. On the upper branch at $S \geq 2.3$ they fall below the momentum root at L=24 — a further reason not to quote an upper "smallest eigenvalue" magnitude.

Consequence for the PINN. The PINN's lower-branch α_1 at S=2.0 (baseline −0.48820, candidate-005 −0.48808) is within 0.9 % of the converged −0.49257. The evaluator's Table 3 magnitude gates compare against values this oracle does not reproduce. The evaluator was left unedited (program-evolution contract); the oracle-v2 numbers are recorded as the recommended reference should a new evaluator version ever be locked.

8 Summary of findings

8.1 What reproduces

- **Table 1 (C_f) and Table 2** (Nu) at the baseline row: upper C_f to 0.004 %, lower C_f to 0.42 %, Nu to 0.24–0.38 % — the latter at the resolution of the independent oracle, which is itself 0.2–0.3 % from the published values.
- **The critical suction parameter** $S_c = 1.849$ against the published 1.8391 (0.54 %).
- **The dual-solution structure and both stability signs**: upper branch stable, lower branch unstable, at every tabulated S and by every method used.
- **Off-search-point behaviour**: three unseen Table 1 rows over three seeds, upper branch and both Nu within 0.34 % of the converged oracle.

8.2 What does not reproduce

- **Table 3 eigenvalue magnitudes.** The lower column is off by factors of 2.8–13; three independent solvers of the paper's own Eq. (31) agree with each other to $< 10^{-7}$ relative and not with the paper. The upper column is reproducible to 0.17–0.70 % only by a boundary-row placement that is mathematically wrong, and the upper magnitude is in any case not identifiable either by this PINN (continuous spectrum for $\alpha > K$) or reliably by the oracle at $S \geq 2.2$.
- **The signs in Table 2.** The article prints negative values under the heading $\mathrm{Nu}\,\mathrm{Re}_x^{-1/2}$ while defining that quantity as $-\theta'(0) > 0$; both the PINN and the independent oracle return positive values of the same magnitude. Verified against page 7; see Section 9.
- **Candidate-003's claimed accuracy gain.** A real CPU-level improvement (0.19 % → 0.018 % lower-C_f error at a 300-iteration cap) that did not survive full-mode evaluation.
- **The session-1 CPU full-mode run.** Never completed; the output file is empty and is documented as empty.

8.3 What was improved

- **2.72× end-to-end same-GPU speed-up** (1389.9 s → 511.6 s) from candidate-005, with accuracy unchanged at the search point and *equivalent* — under a pre-registered rule — on unseen rows and seeds. Stage-level: 2.48× (Table 1/2), 2.45× (fold), 3.13× (stability).
- The dominant mechanism is **replacing three nested autograd passes with closed-form Taylor-mode derivative jets**, verified gradient-identical to 2×10^{-14} before evaluation; the rest comes from adaptive LBFGS stopping and a cheaper L_{MAP} pilot.

- A **corrected, converged Table 3 oracle** now exists where none did, including the previously unexamined energy-disturbance spectrum.

9 Did this work beat the article?

The honest verdict is **partial, and the partition is sharp**: this work *supersedes* the article on its stability table and provides a better numerical reference for its flow tables, while the article remains ahead on raw computational cost and on breadth of parameter coverage. The two claims should not be traded against each other, and the speed figure reported in Section 6 is *not* a claim against the paper's method.

Table 10: Axis-by-axis comparison against Pandey *et al.*

Axis	**The article**	**This work**	**Verdict**
Stability eigenvalue magnitudes (Table 3)	Published to 7 decimals; not recoverable from its own Eq. (31)	Converged values by three independent methods, agreeing to $< 10^{-7}$	**superseded**
Energy-disturbance spectrum (Eq. 32)	Not examined	Computed; all eigenvalues positive, no sign change	**new**
Far-field reference for Tables 1/2	$\eta_\infty = 10$ truncation	Converged $\eta_\infty = 40$ oracle	**improved**
Table 1/2 values themselves	8 significant figures	PINN within 0.004–0.42 %	matched, not beaten
Internal consistency of Table 2	Signs contradict the paper's own definition	Inconsistency identified and verified	**defect found**
Compute per parameter point	RK4 + secant: milliseconds	511.6 s for the full three-stage pipeline	**article wins**
Parameter coverage and physical interpretation	Dense grids, 14 figures, Section-6 synthesis	One gated row, S_c, S=2.0 stability, 3 unseen rows	**article wins**
Auditability of the record	Printed tables	Hashes, locks, pre-registrations, raw sweeps, negative results	**this work**

9.1 Where this work supersedes the article

Table 3 is the one place where the article's published numbers are shown to be wrong rather than merely imprecise. Three independent solves of the paper's *own* Eq. (31) — IVP shooting with $g'(L) = 0$, a corrected Chebyshev discretisation, and shooting with the exact asymptotic boundary condition — agree with one another to $< 10^{-7}$ relative at every tabulated S and change by $< 10^{-5}$ between L=20 and L=24, while differing from the published lower-branch column by factors of 2.8 to 13 (Table 9). No correct solver can land within 1 % of -0.0376726 at S=2.0. The qualitative conclusion the article draws from Table 3 — upper branch stable, lower branch unstable at every S — is unaffected: every method reproduces both signs. What does not survive is the quantitative column.

Two further contributions have no counterpart in the article: a mechanical candidate explanation for how the published *upper* column could have arisen (a boundary-row placement that is mathematically wrong but reproduces all five values to within 0.70 %), and the first computation of the energy-disturbance spectrum of Eq. (32), which turns out to be everywhere positive and therefore to change neither sign conclusion — while also showing that on the upper branch at $S \geq 2.3$ it falls *below* the momentum root, one more reason the article's upper magnitudes should not be quoted.

9.2 Where this work improves the article's numbers without overturning them

The article truncates the far field at $\eta_\infty = 10$. The converged $\eta_\infty = 40$ oracle built here shows what that costs on the slowly-decaying lower branch: the published lower C_f is 0.17 % below the converged value at S=1.95 and 0.78 % below it at S=2.1, with the gap growing as the branch decays more slowly. On Nu the oracle sits 0.20 % (upper) and 0.29 % (lower) from the published values. The practical consequence is that the article's own tables — not the PINN — are the limiting reference: the PINN's 0.24–0.38 % agreement on Nu is at the resolution of the best available independent solve, so no finer comparison is meaningful. The PINN itself is therefore a faithful reproduction, not an improvement, of Tables 1 and 2; the *oracle* is the improvement.

9.3 A verified internal inconsistency in Table 2

The article defines $\mathrm{Nu}\,\mathrm{Re}_x^{-1/2} = -\theta'(0)$ (Section 3), and with $\theta(0) = 1$, $\theta(\infty) = 0$ this is positive for the cooling wall considered. Page 7, Table 2, nevertheless prints *negative* values under the heading $\mathrm{Nu}\,\mathrm{Re}_x^{-1/2}$: the baseline row ($S$=1.95, K=0.1, β=0.1, Pr=2, A^*=B^*=0.1) reads −3.35807824 and −3.14427924, and every row in the table carries the same sign. Both the PINN and the independent `scipy.solve_bvp` oracle return +3.350 and +3.132 — magnitudes that agree to 0.2–0.4 %. The table therefore appears to tabulate $\theta'(0)$ while labelling it $-\theta'(0)$; it is a presentational slip, not a numerical error, and the figures on the same page (Figs. 9–10, plotted on positive axes) are consistent with the positive convention. It is recorded here because a reader copying Table 2 verbatim would carry the wrong sign into downstream work. The evaluator gates on magnitude, so nothing in this project is affected. This was verified directly against page 7 of the archived copy whose SHA-256 is recorded in Table 1.

9.4 Where the article still wins

- **Computational cost.** The article's shooting method (RK4, $\Delta\eta = 0.01$, secant to 10^{-5}) solves a branch in milliseconds. The fastest program produced here needs 511.6 s for the full three-stage pipeline. The 2.72× reported in Section 6 is measured against the supplied PINN baseline on identical hardware and says nothing about the paper's own solver; a PINN was never going to win that comparison, and nothing in this record claims it does. The case for the PINN is that it imposes the far-field conditions exactly, passes smoothly through the turning point, and provides a third independent route to the same claims — not that it is cheap.
- **Breadth.** The article sweeps dense grids in S, K, β, Pr, A^* and B^*, with fourteen figures and a synthesis of the physical mechanisms. This project gates one parameter row, plus S_c, plus the S=2.0 stability row, with three unseen rows over three seeds as confirmation. The qualitative conclusions of the article's Section 6 were transcribed as constraints to protect, never independently re-derived across the full grid.
- **Physical interpretation.** No claim about the physics of the Maxwell shrinking-sheet problem originates here.

9.5 What would actually constitute beating it

Two things, neither yet done:

1. **Publish the Table 3 correction.** The evidence is complete and preserved — three agreeing methods, domain-length studies, pre-registered criteria applied mechanically, and a candidate attribution for the published upper column. What is missing is the write-up and the courtesy of putting it to the authors; the article is an uncorrected proof, so the window for a correction may still be open.
2. **Execute the D-track** (Section 4.3, `candidate-mechanisms.md`). A single parametric network $f(\eta; S, \beta, K)$, $\theta(\eta; S, \beta, K, \mathrm{Pr}, A^*, B^*)$ would reproduce the entire Table 1/2 sweep from one

training run and extend past it — including the non-existence region below S_c, which the article maps only by where its solver stops converging. Differentiating that network with respect to S gives continuous fold-tracking and a continuous $\alpha_1(S)$, which would verify as *one* claim that the turning point and the stability-exchange point coincide. The article asserts this through two separate tables and never checks it. That is the result that would be beyond the paper rather than alongside it, and it needs held-out (S, β, K) points and its own evaluator, not a candidate folder.

10 Methodological discipline actually followed

This is a large part of the value of the record and is worth stating explicitly:

1. **Pre-registration before results.** Three pre-registration documents (`GPU-SEARCH-PREREGISTRATION.md`, `OPEN-ITEMS-PREREGISTRATION.md`, and the `oracle-v2` criteria mechanically applied by `judge.py` into `judge.md`) fix acceptance rules, thresholds and convergence criteria in advance. When an outcome fell outside the foreseen cases (Item B upper branch), that is reported as such rather than reinterpreted.
2. **The evaluator was never edited**, not even when two of its gates were proven unmeetable by any correct solver. The gates are reported, excluded from acceptance with a documented reason, and the corrected reference values are recorded for a future locked evaluator version.
3. **No cross-hardware timing claims.** CPU and GPU numbers are reported separately and never summed; session-3 timings are never compared with session 2's because the GPU differs; the 1.85× fresh-row ratio is explicitly marked as not comparable with the 2.72× full-mode figure.
4. **Hash-verified reconstruction.** The winner was rebuilt from the locked baseline plus its recorded patch and verified byte-identical before confirmation.
5. **Paired, idle-machine, serial measurement** with recorded timelines, `nvidia-smi` output, environment dumps and SHA-256 manifests next to every run.
6. **Smoke runs are never reported as reproduction.** Stated in the evaluator docstring, the project notes, and the skill.
7. **Negative and null results are kept** — candidate-003's non-reproducing gain, candidate-004's within-noise speed-up, the dead CPU run, the refuted spectrum hypothesis.
8. **Honest statistical limits.** "Equivalent" is defined as "no gross difference detected at $n = 9$", not as equality; the non-pairing of seeds across the two programs is noted.
9. **Source material untouched.** Zero modified files against upstream; all new work in two clearly separated directories.

11 Evidence inventory

Table 11: Principal artifacts under `initiatives/maxwell-shrinking-pinn/projects/pinn-reproduction/`.

Path	Content
`../BRIEF.md`	Original request, provenance hashes, desired outcome, constraints
`NOTES.md`	Methodology, per-session status, limitations, reproduction commands

Path	Content
paper/paper-summary.md	Transcribed equations, BCs, Tables 1–3 with gate markings, Section-6 conclusions
paper/pagination_FINMEC_100352-1.pdf	Read-only copy of the source PDF
baseline/pinn_maxwell_shrinking.py	Unmodified three-stage PINN (the program to evolve)
oracle/reference_bvp.py	Independent `scipy.solve_bvp` ground truth (unmodified)
oracle/ref_eig_cheb.py	Independent Chebyshev eigenvalue solver (unmodified; BC rows at swapped ends)
evaluator.py	Candidate-proof evaluator; smoke/full modes, `-fold`, `-stability`, `-oracle`
baseline-lock/, baseline-lock-gpu/	Hash-locked baseline + evaluator snapshots, CPU and GPU
candidates/candidate-001..005/	Per-hypothesis folder: `NOTE.md`, `patch.diff`, full source, `smoke.json`, `full.json`, stderr logs
candidates/LINEAGE.md	Lineage table with timings, speed-ups and gate outcomes
runs/full_confirmation_gpu.json	Same-GPU full-mode reference (1451.8 s)
runs/GPU-SEARCH-PREREGISTRATION.md	Acceptance rule and the two non-identifiability arguments
runs/GPU-SEARCH-RESULTS.md	Winner confirmation, paired timings, fresh-parameter check, verdict
runs/confirm-005/	Paired baseline/winner runs, fresh-seed run, `env.txt`, `nvidia-smi.txt`, `timeline.txt`, run script
runs/OPEN-ITEMS-PREREGISTRATION.md	Session-3 rules for Items A and B, with program hashes
runs/fresh-paired-2060s/ANALYSIS.md	Item A result: per-point table, means/maxima, verdict, cross-GPU note
runs/fresh-paired-2060s/analyze.py	The analysis script that produced it
runs/full_table12_confirmation.json	Session-1 CPU run that never completed (empty, documented)
confirm_fresh.py	Unseen-row, multi-seed confirmation harness
diagnostics/eig_shooting.*, eig_cheb_fixed.*	Independent eigenvalue solves with domain-length studies (code + output)
oracle-v2/	Corrected Table 3 oracle: solver, raw JSON, `judge.py`/`judge.md`, `README.md` with the full table
../../../skill-lab/ evolve-maxwell-pinn-gpu/	`SKILL.md` and `references/candidate-mechanisms.md` (14 mechanisms, A–D)

11.1 Commit history

The 6.7 GB GPU venv is excluded from version control; `open-discovery`'s own upstream history is preserved as `open-discovery/.git-upstream` (rename back to `.git` to restore it as a standalone checkout).

12 How to reproduce

```
# environment (GPU; keeps sm_61 kernels for a GTX 1080 Ti)
uv pip install -r amit-pinn-article/requirements-gpu-cu126.txt \
  --extra-index-url https://download.pytorch.org/whl/cu126
```

Commit	Date	Subject
9ee93bc	2026-11-05	Maxwell shrinking-sheet PINN: Open Discovery GPU reproduction and candidate search
d5fd1ba	2026-11-05	Confirm candidate-005: 2.72× same-GPU speed-up; record results and lineage
2e1be62	2026-11-06	Session 3: pre-register open items; Table 3 oracle v2; start paired fresh-seed runs
68b53ec	2026-11-06	Sync: paired fresh-seed run progress (baseline seeds on 2060 SUPER)
4385559	2026-11-06	Sync: candidate-005 fresh-seed run progress (2060 SUPER)
30c62ea	2026-11-06	Close open items: baseline vs 005 fresh-seed spread equivalent; Table 3 oracle v2 results

```
cd amit-pinn-article/open-discovery/initiatives/maxwell-shrinking-pinn/\
projects/pinn-reproduction

# cheap pipeline sanity gate (~15 s GPU, ~127 s CPU)
python evaluator.py baseline/pinn_maxwell_shrinking.py

# the actual paper-accuracy claim (~1450 s on a 1080 Ti for all three stages)
python evaluator.py baseline/pinn_maxwell_shrinking.py \
  --mode full --oracle --fold --stability

# unseen-row, multi-seed confirmation
python confirm_fresh.py baseline/pinn_maxwell_shrinking.py 1 2 3

# rebuild the winner from the locked baseline + its recorded patch
cp baseline-lock-gpu/baseline/pinn_maxwell_shrinking.py /tmp/c005.py
patch /tmp/c005.py < candidates/candidate-005/patch.diff

# corrected Table 3 oracle (~25 min, CPU)
python oracle-v2/table3_oracle.py && python oracle-v2/judge.py

# independent eigenvalue diagnostics
python diagnostics/eig_shooting.py
python diagnostics/eig_cheb_fixed.py

# re-lock after any intentional change to baseline/ or evaluator.py
python ../../../../program-evolution/scripts/lock_baseline.py \
  --baseline baseline/pinn_maxwell_shrinking.py --evaluator evaluator.py \
  --goal "<state the goal>" --output baseline-lock-v2 --timeout-seconds 1800
```

13 Status, limitations and possible next steps

Status. Reproduction complete to the extent the paper allows, with no open items. Tables 1 and 2 and S_c reproduce; Table 3 signs reproduce and magnitudes do not, with the discrepancy localised to the paper side and evidenced by three independent solvers. Candidate-005 is confirmed 2.72× faster on the same GPU and accuracy-equivalent to the baseline.

Limitations to carry forward.

- The evaluator still carries two Table 3 magnitude gates that no correct solver can pass, so `valid=false` on *every* full-mode run including the untouched baseline. Any future locked evaluator version should adopt the oracle-v2 reference values.
- The lower-branch C_f shows a $\sim$0.03–1.0 % seed spread against the converged oracle. This is inherent to the baseline architecture, not to any candidate, and is partly GPU-level nondeterminism ($\sim$0.5 pp at identical seeds across devices).
- No reliable upper-branch α_1 magnitude exists at $S \geq 2.2$ by any method used here.
- All accuracy gates are at a single parameter point; $n = 9$ off-search-point checks detect only gross differences.
- Mechanisms A1 (batching both branches via `vmap`), A4 (Adam/LBFGS rebalance), A5 (vectorising the fold continuation loop) and A6 (larger collocation batches) from the mechanism menu remain untried.

Possible next steps. The D-track in `references/candidate-mechanisms.md` is genuinely new science beyond the paper and would require a *new project* with its own evaluator and held-out (S, β, K) grid points: a parametric/meta-PINN $f(\eta; S, \beta, K)$ reproducing the entire Table 1/2 sweep in one training run (and mapping the non-existence region below S_c); continuous fold-tracking by differentiating that network with respect to S; parametric stability producing a continuous $\alpha_1(S)$ and locating $\alpha_1 = 0$ directly, which would check as *one continuous claim* that the turning point and the stability-exchange point coincide — something the paper asserts through separate tables but never verifies; and a small multi-seed ensemble reporting C_f/Nu spread as a robustness companion.

A note on scope. This initiative is a reproduction plus harness-and-optimisation study. It asserts no new claim about the physics of the Maxwell shrinking-sheet problem. The one finding with implications beyond the harness — that Table 3's eigenvalue magnitudes are not recoverable from the paper's own Eq. (31), while a specific incorrect boundary-row placement does recover its upper column to within 0.7 % — is stated as evidence, with the raw sweeps, three independent methods, and pre-registered criteria preserved so a reader can judge it independently.